Ex 15 Robot tank on line

2) The goal here is not to follow a moving point with time, but a specific heading θ_d.

The state vector is :
$$\begin{cases} \dot{x} = v\cos\theta \\ \dot{y} = v\sin\theta \\ \dot{\theta} = u_2 \\ \dot{v} = u_1 - v \end{cases}$$

We could make a feedback linearization if it was to follow a specific target point $\begin{pmatrix} x_d \\ y_d \end{pmatrix}$

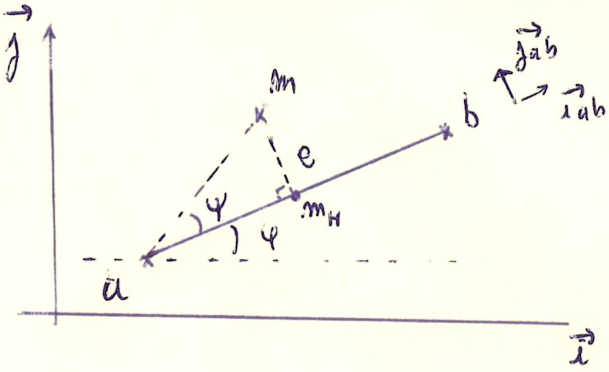
But we could just simply use $u_2 = \theta_d - \theta$ because $\dot{\theta}_d = 0$ means θ is steady and equal to θ_d.

As the lecture, in order to have $\theta_d - \theta$ modulo 2π and come back to 0, we use a sawtooth function like $\arctan\left(\tan\left(\frac{\theta_d - \theta}{2}\right)\right)$

To have a smooth acceleration, we use $\tanh(v_d - v)$ for u_1

So, $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} a_1 \cdot \tanh(v_d - v) \\ a_2 \cdot \text{sawtooth}(\theta_d - \theta) \end{pmatrix}$ where v_d is the target speed and a_1, a_2 some constant to define.

3)



We need to reduce error e which is the distance between m and the line ab.

Then, proportionnaly substract e to the target φ when m is far.

So we choose $\theta_d = \varphi - \arctan(e)$ if e is big. So we will always add an angle in $]-\frac{\pi}{2}, \frac{\pi}{2}[$. e should be signed, to know if we are under or above the line ab. We choose to project $\vec{am}$ onto $\vec{j}_{ab}$ to get e:

$$e = \vec{am} \cdot \vec{j}_{ab} = \begin{pmatrix} x - x_a \\ y - y_a \end{pmatrix} \cdot \begin{pmatrix} -y_{\vec{i}_{ab}} \\ x_{\vec{i}_{ab}} \end{pmatrix} = \begin{pmatrix} x - x_a \\ y - y_a \end{pmatrix} \cdot \begin{pmatrix} -(y_b - y_a) \\ x_b - x_a \end{pmatrix} \cdot \frac{1}{\|\vec{ab}\|}$$

4) When a robot finishes tracking a line, ~~with~~ we increment the line coordinates.

5) To give different speed for each robot, we simply give a vector V_d of desired v_d target for each robot.

To avoid collisions, we choose to wait that ~~wait~~ the robot above finish its line to launch the following robot.